KARDELEN SURAL

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Research Interests

Macroeconomics, Labor Economics, Technological Change and Automation, Income Distribution

Education

- **Carnegie Mellon University, Tepper School of Business** *2026 – Present*
  Ph.D. Program in Economics

- **Bilkent University** *2024 – 2026*
  M.A. in Economics
  Advisor: A. Arda Gitmez
  Thesis: *Unions in the Age of Automation*

- **Bilkent University** *2019 – 2024*
  B.A. in Economics
  Dean's High Honor Roll
  Thesis: *Bridging the Gap: Navigating Income Inequality in Türkiye and Argentina through Crisis and Recovery*
  First Place, Transdisciplinary Senior Project, Bilkent University

Research

- **Unions in the Age of Automation (with Arda Gitmez and Özlem Kına)** *Work in progress*
  Develops a search-and-matching model in which firms' automation incentives interact with workers' unionization decisions. At intermediate productivity levels, the threat of automation deters unionization and suppresses wages despite little realized automation; at high productivity, unionization and labor-market opportunities recover. Evaluates automation caps and union-provided employment protection as policy responses.

- **Bridging the Gap: Navigating Income Inequality in Türkiye and Argentina through Crisis and Recovery** *2024*
  Undergraduate thesis, Bilkent University. Examines changes in income inequality surrounding the Global Financial Crisis in Türkiye and Argentina and studies the role of minimum wages, taxation, and policy responses using empirical analysis in R. Awarded first place in Bilkent University's Transdisciplinary Senior Project competition.

Research Experience

- **Graduate Researcher, Carnegie Mellon University** *2026 – Present*
  Research assistance for Stephen Spear and Selman Erol on the paper *Optimal Bureaucracy.*

- **Graduate Researcher, Bilkent University** *2024 – 2026*
  Advisor: A. Arda Gitmez
  Developed *Unions in the Age of Automation.*

- **Research Assistant to Assistant Prof. Özlem Kına, Bilkent University** *2026*
  Verified and checked mathematical proofs for the paper "Optimal Taxation of Labor-Replacing Technologies." Reviewed theoretical derivations for mathematical consistency and correctness.

- **Research Assistant to Prof. Dr. H. Cagri Saglam, Bilkent University** *2022 – 2025*
  Conducted research on infrastructure and economic growth; contributed to game-theoretic modeling of power transitions and preventive war in "A Game-theoretic Investigation of the Thucydides Trap" with Karagözoglu and Keskin; conducted literature reviews on status-seeking and habit formation under reference-dependent preferences; and contributed to research on the economics and evolution of sleep.

Teaching Experience

- **Instructor, Bilkent University** *July 2026*
  Designed and taught a course on Economic Growth and Income Distribution high school students. Introduced students to core economic concepts and guided small-group discussions and exercises.
- **Teaching Assistant, Bilkent University** *2021 – 2026*
  Courses: Macroeconomic Theory II (ECON 206), Mathematics for Economists (ECON 225), Calculus I–II (MATH 105–106).
  Led recitations and problem-solving sessions, held office hours, and graded assignments and examinations.

Selected Coursework

- **Mathematics:** Real Analysis/ Advanced Calculus, Linear Algebra, Differential Equations, Finite and Discrete Mathematics, Mathematical Foundations of Data Science
- **Economics:** Mathematical Economics, Advanced Economic Theory, Macroeconomic Theory I–II, Political Economy, Theories of Growth and Development, Graph Theory in Economics

Honors and Awards

- Outstanding Talent Medal, Presidency of Türkiye *2025*
- Best Teaching Assistant Award for my work in ECON206 (Macroeconomic Theory II), Bilkent University *2025*
- TÜBİTAK BİDEB National MSc/MA Grant *2024*
- First Place, Transdisciplinary Senior Project, Bilkent University *2024*
- Comprehensive Scholarship, Bilkent University *2019 – 2026*
- Outstanding Success Scholarship, Turkish Education Foundation (TEV) *2019 – 2024*

Internships

- **Intern, Ministry of Culture and Tourism** *2023*
  Department of Strategic Planning and Coordination.
- **Intern, Ministry of Treasury and Finance** *2022*
  Directorate of Financial Markets. Prepared reports on financial instruments, foreign-exchange policies, and access to credit.

Test Scores

- GRE Quantitative: **170/170** *2025*
- Academic Personnel and Graduate Education Entrance Exam (ALES): **94.19/100**; ranked 699th among 158,446 candidates (**top 0.44%**) *2024*

- National University Entrance Examination: ranked 213th among approximately 2.5 million candidates (**top 0.0085%**) *2019*
- IELTS Academic: **7.5**

Technical Skills

- **Programming and Statistical Software:** MATLAB, Python, R, Stata
- **Typesetting and Productivity:** LaTeX, Microsoft Office